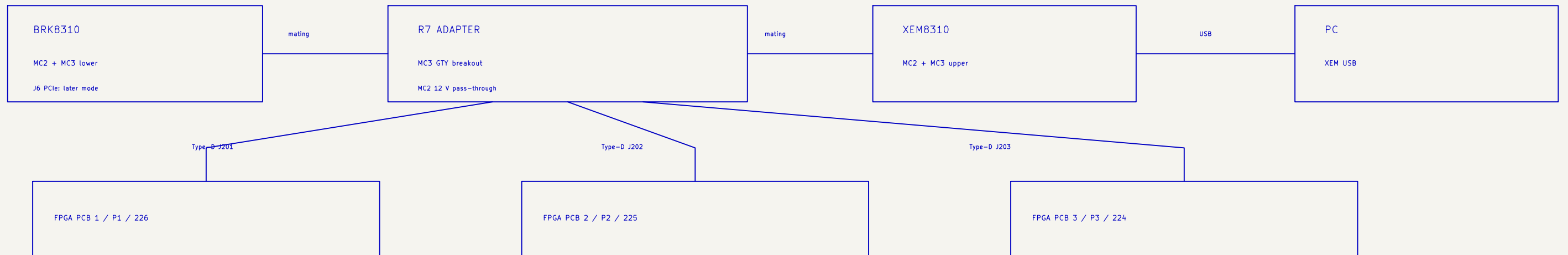


R7 CANDIDATE | THREE SIMULTANEOUS CUSTOM FPGA LINKS

Physical stack and cable fan-out. Four data pairs per cable; each port has a separate JTAG and power path.



02 / PORT 1 - MC3 BANK 226 - TYPE-D J201

03 / PORT 2 - MC3 BANK 225 - TYPE-D J202

04 / PORT 3 - MC3 BANK 224 - TYPE-D J203

05 / MC2 POWER + MC3 BANK-67 GPIO

SIGNAL CHOICE

MC3 has 12 GTY channels in banks 224/225/226. Each port uses RX0/RX1/RX2 + TX0 in one bank.

Selected XC7A25T: recording pair 2 is physically wired but reserved. TX0 control shares channel 0.

Selected XEM GTY contacts go only to cables; matching BRK contacts are open. J6 PCIe is unavailable in this mode.

IMPLEMENTATION BOUNDARY

All MC1 contacts 1–80 and MC2 contacts 9–80 require pass-through; not netlisted. MC3 others are shown.

Connector footprints, MC3 ground paddle, orientation, 5 mm stack changes, bracket/PCIe fit and SI: TBD.

Protection blocks and three probe headers are drafts. XEM-to-target JTAG bridge and gateway: not implemented.

No PCB or current, cable, firmware, BER, USB and simultaneous-capture qualification. NOT FOR FAB.

Custom Type-D carries proposed 12 V on contact 19; never connect to ordinary HDMI equipment.

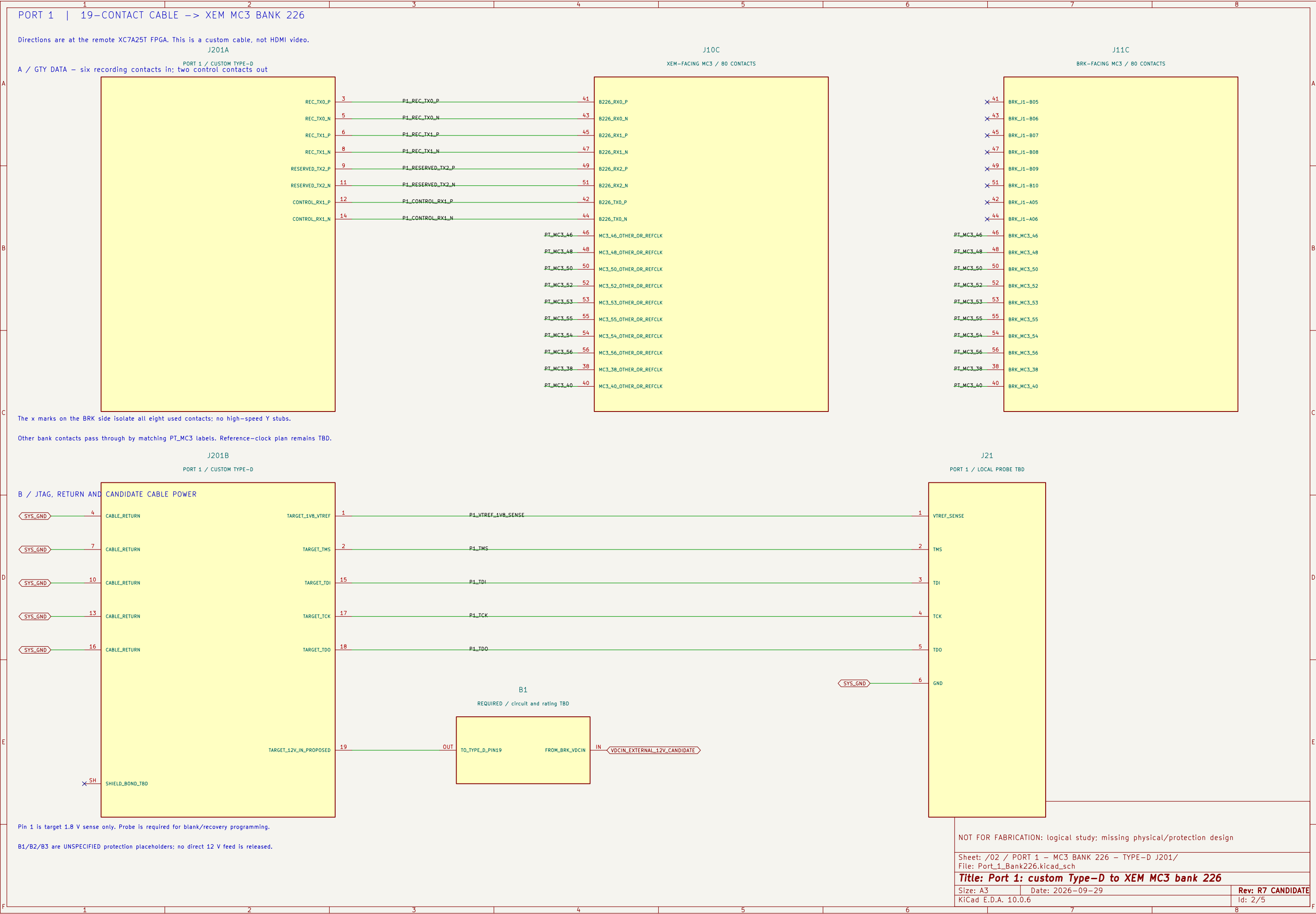
NOT FOR FABRICATION: logical study; missing physical/protection design

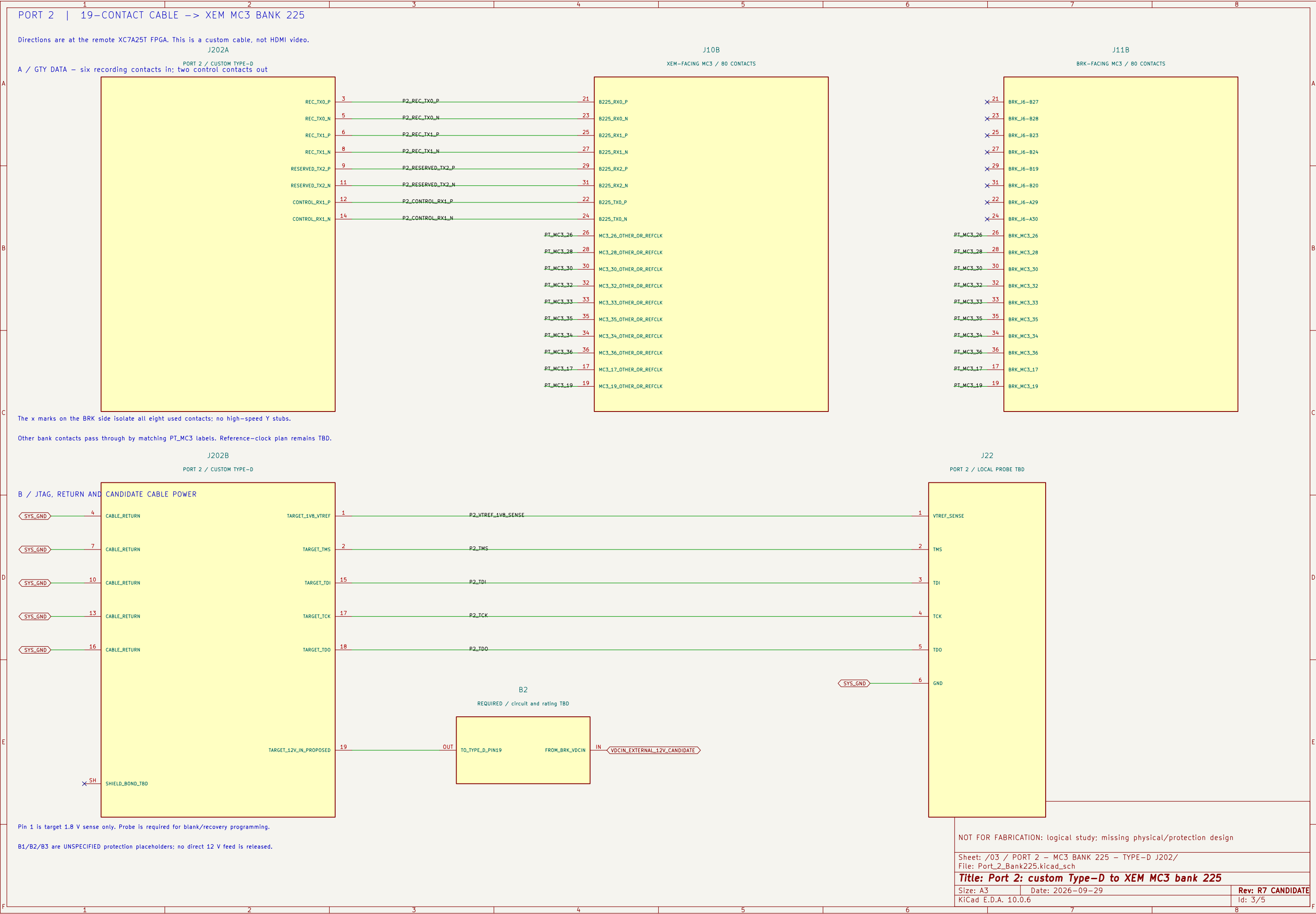
Sheet: /
File: BRK8310_Three_Port_Adapter_R7_Candidate.kicad_sch

Title: XEM8310 / BRK8310 three-cable adapter - circuit map

Size: A3	Date: 2026-09-29
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KiCad E.D.A. 10.0.6	Id: 1/5
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The x marks on the BRK side isolate all eight used contacts; no high-speed Y stubs.

Other bank contacts pass through by matching PT_MC3 labels. Reference-clock plan remains TBD.

J202B

PORT 2 / CUSTOM TYPE-D

B / JTAG, RETURN AND CANDIDATE CABLE POWER

SYS_GND

SYS_GND

SYS_GND

SYS_GND

SYS_GND

4

7

10

13

16

CABLE_RETURN

CABLE_RETURN

CABLE_RETURN

CABLE_RETURN

CABLE_RETURN

TARGET_1V8_VTREF

TARGET_TMS

TARGET_TDI

TARGET_TCK

TARGET_TDO

1

2

15

17

18

P2_VTREF_1V8_SENSE

P2_TMS

P2_TDI

P2_TCK

P2_TDO

VTREF_SENSE

TMS

TDI

TCK

TDO

GND

SH

SHIELD_BOND_TBD

B2

REQUIRED / circuit and rating TBD

TO_TYPE_D_PIN19

FROM_BRK_VDCIN

OUT

IN

VDCIN_EXTERNAL_12V_CANDIDATE

J22

PORT 2 / LOCAL PROBE TBD

Pin 1 is target 1.8 V sense only. Probe is required for blank/recovery programming.

B1/B2/B3 are UNSPECIFIED protection placeholders; no direct 12 V feed is released.

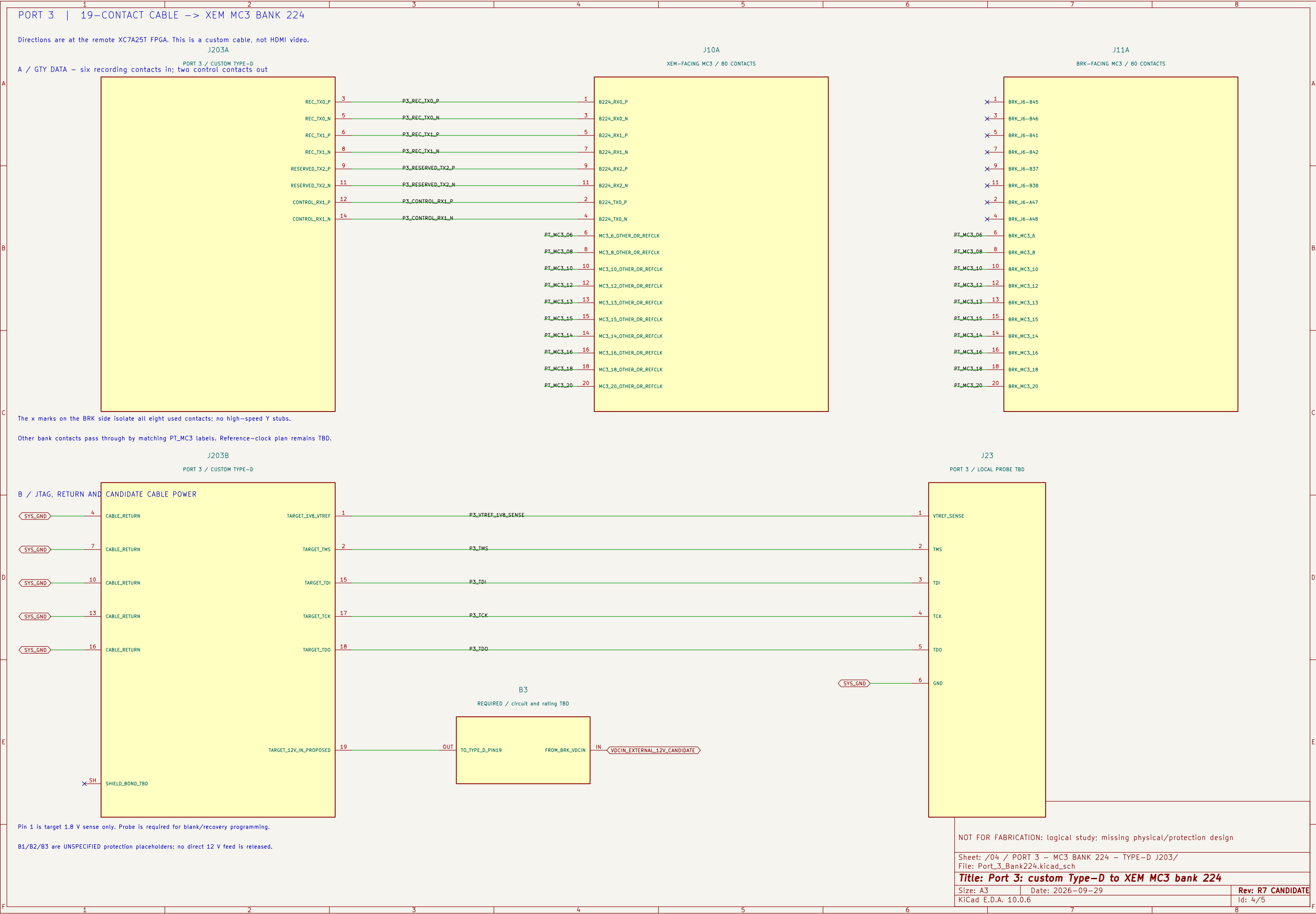
NOT FOR FABRICATION: logical study; missing physical/protection design

Sheet: /03 / PORT 2 - MC3 BANK 225 - TYPE-D J202/
File: Port_2_Bank225.kicad_sch

Title: Port 2: custom Type-D to XEM MC3 bank 225

Size: A3Date: 2026-09-29Rev: R7 CANDIDATE

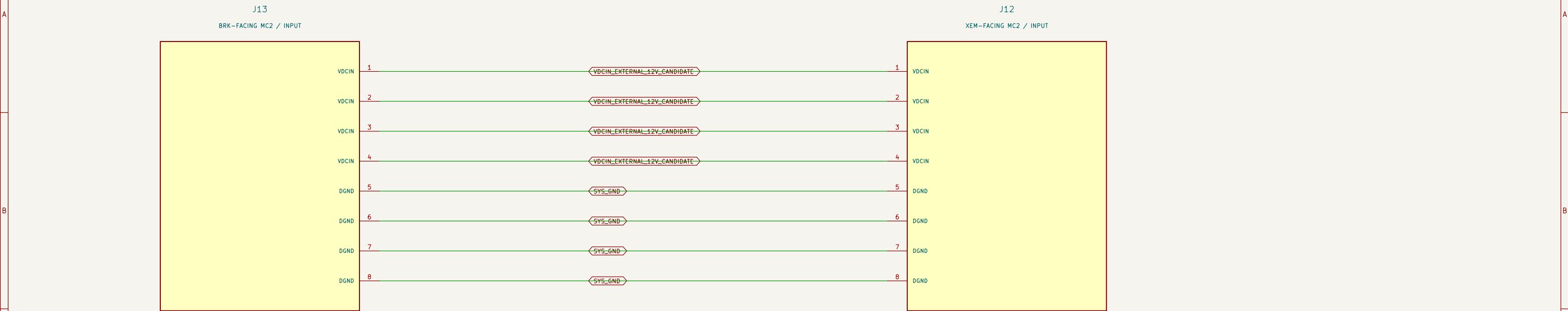
KiCad E.D.A. 10.0.6Id: 3/5



1	2	3	4	5	6	7	8
1. NO. POWER 1. EXTERNAL 1. IN THROUGH BEM TO MEN							

The XEM does not generate 12 V. Its VDCIN contacts receive the external BRK supply.

MC2 contacts 1-4 = VDCIN; 5-8 = DGND. These eight contacts are explicit here.



BRK power input and the three 12 V branch loads require a common current/fault budget.

Separate branch protection, cable/contact current, inrush and remote regulators: unresolved.

All MC1 contacts 1–80 and MC2 contacts 9–80 require pass-through; R7 does not netlist them.

MC3 BANK-67 GPIO | BRIDGE RESERVATION J10D J11D



57/59/61/63/65: isolated reservations for a possible XEM JTAG bridge.				
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Bank 67 is 1.2 V. No translator, source select or gateway is implemented.

NOT FOR FABRICATION: logical study; missing physical/protection design

Sheet: /05 / MC2 POWER + MC3 BANK-67 GPIO/
File: MC2_Power_and_MC3_GPIO.kicad_sch

Title: MC2 input power and MC3 bank-67 GPIO

Size: A3	Date: 2026-09-29	Rev: R7 CANDIDATE
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Size: A3	Date: 2026-09-29	Rev: R7 CANDIDATE
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Size: A3	Date: 2026-09-29	Rev: R7 CANDIDATE
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KiCad E.D.A. 10.0.6	Id: 5/5
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KiCad E.D.A. 10.0.6	Id: 5/5
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